

REPUBLIC OF CAMEROON



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MINISTRY OF HIGHER EDUCATION

INSTITUT UNIVERSITAIRE DE LA
COTE

REPUBLIQUE DU CAMEROUN



Paix-Travail-Patrie

MINISTRE DE L'ENSEIGNEMENT
SUPERIEUR

INSTITUT UNIVERSITAIRE DE LA
COTE



INTERNSHIP REPORT

Design and implementation of a web platform
for the integration and technical training of
trainees at ENEO Cameroon.

Internship done from 7th of July to 31th of August 2025 in partial fulfillment of the HND (Higher National Diploma)

Field of study: **Computer Engineering**

Specialty: **software engineering**

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Under the supervision of:

PROFESSIONAL SUPERVISOR

Mr. NYALEU PATRICK

Academic Year
2024-2025

CERTIFICATION

This is to certify that an internship with the theme: **“DESIGN AND IMPLEMENTATION OF A WEB PLATFORM FOR THE INTEGRATION AND TECHNICAL TRAINING OF TRAINEES AT ENEO CAMEROON.”**: case of **Internes_Eneo**, is

Originally written and presented by **KOUAKEP KAMTO JEREMIE OTHNIEL** after a two month of internship carried out at ENEO CAMEROON to meet up with the partial requirement and regulation governing the award of the Higher National Diploma (HND) of Institut Universitaire de la Cote in the field software engineering.

DEDICATION

TO MY FAMILY

ACKNOWLEDGEMENTS

The accomplishment of this work would not have been possible without the support and contributions of several people. Although it is difficult to mention everyone individually, I extend my most sincere gratitude to all who played a part in this achievement.

- To all the lecturers and professionals who have contributed to the knowledge instilled in me throughout my academic journey.
- Mr. YVES NANFACK, for supervising me during my internship at ENEO Cameroon, and also for his guide and advices during my internship.
- To **Mr. Patrick, Mr. Steve, Mme. Magali, and Mme. Emanou**, for their support, guidance, and assistance during my internship.
- To **ENEO Cameroon**, for granting me the opportunity to undertake this internship and for providing a conducive learning environment.
- **MY LOVELY AND SUPPORTIVE FAMILY** for all their moral and financial support especially for the furthering my education.
- All my **FRIENDS AND CLASSMATES** who contributed in one way or the other for the realization of this project.
- To everyone I have failed to mention here, thank you.

ABSTRACT

The integration and technical training of trainees is a crucial activity for companies such as ENEO Cameroon, as it ensures the smooth transfer of knowledge, the development of professional skills, and the preparation of future experts in the energy sector. With the continuous evolution of technology and the growing need for efficiency, the traditional methods of managing trainees — from registration to training follow-up — have become limited and sometimes cumbersome.

The advancements in web technologies now make it possible to design platforms that simplify and automate these processes, with emphasis on reliability, security, and ease of use. Such a system must allow the creation and management of trainee accounts, the submission of internship-related documents, the follow-up of training programs, and the facilitation of communication between trainees and supervisors.

In this context, we have undertaken the **design and implementation of a web platform dedicated to the integration and technical training of trainees at ENEO Cameroon**. The objective is to provide a user-friendly and secure solution that ensures effective management of trainee data, promotes better supervision, and contributes to the success of ENEO's capacity-building mission.

RESUME

L'intégration et la formation technique des stagiaires représentent une activité essentielle pour une entreprise comme **ENEO Cameroun**, car elles garantissent le transfert des connaissances, le développement des compétences professionnelles ainsi que la préparation des futurs experts du secteur de l'énergie. Avec l'évolution constante des technologies et la nécessité croissante d'efficacité, les méthodes traditionnelles de gestion des stagiaires — allant de l'inscription au suivi de la formation — se révèlent limitées et parfois contraignantes.

Les avancées dans le domaine des technologies web offrent désormais la possibilité de concevoir des plateformes qui simplifient et automatisent ces processus, tout en mettant l'accent sur la fiabilité, la sécurité et la convivialité. Un tel système doit permettre la création et la gestion des comptes stagiaires, la soumission des documents liés au stage, le suivi des programmes de formation, ainsi que la facilitation de la communication entre stagiaires et encadreur.

C'est dans ce contexte que nous avons entrepris la **conception et la mise en œuvre d'une plateforme web dédiée à l'intégration et à la formation technique des stagiaires au sein d'ENEO Cameroun**. L'objectif est de proposer une solution ergonomique et sécurisée, assurant une gestion efficace des données des stagiaires, favorisant un meilleur encadrement et contribuant à la réussite de la mission de renforcement des capacités menée par ENEO.

PREFACE

With the aim of ensuring a suitable development and providing business with competent and competitive workforce in various fields, the Cameroon government through the ministry of higher education has opened the private institution of education. This then created the HIGER INSTITUTE OF TECHNOLOGY AND INDUSTRIAL DESIGN (ISTDI) by decree N° 02/0094/MINESUP/DDES/ESUP of September 13 2002 and authorization of opening N°01/02/MINESUP/DDES/ESUP of 18 September 2002, located in the coastal region, Wouri department, district of Douala 5th, in Logbessou district. The ISTDI then created the UNIVERSITY INSTITUTE OF THE COAST by decree N° 5/05156/N/MINESUP/DDES/ESUP/SAC/EBM. University Institute of the Coast (IUC) consist of five (5) different institution which are:

❖ *School of Engineering and Applied Science*

It consists of:

- HND INDUSTRIAL PROGRAMS
 - Electrical power systems (EPS)
 - Industrial Computing and Automation (ICA)
 - Telecommunication (TEL)
 - Software Engineering (SWE)
 - Network Security (NWS)
 - Chemical Manufacturing (CM)
 - Mechanical Manufacturing (MEM)
 - Building Science and Technology (BST)

- HND COMMERCIALS PROGRAMS

- Marketing (MKT)
- Management (MGT)

- Accountancy (ACC)
- Banking and Finance (BF)
- Logistic and Transport Management (LTM)
- BTS PROGRAMS
 - GENIE CIVILE (GC)
 - GEOMETRE TOPOGRAPH (GT)
 - BATIMENTS
 - TRAVEAUX PUBLIC (TPU)
- BACHELORS OF ENGINEERING (B-Eng)
 - Computer Science Engineering
 - Electromechanical Engineering
 - Civil and Environmental Engineering
- BACHELORS OF TECHNOLOGY (B-Tech)
 - Chemical Engineering
 - Electrical and Electronics Engineering
 - Civil Engineering
 - Software Engineering
 - Mechatronics Engineering
 - Industrial Automation and Computing
 - Welding and Fabrication Technology
 - Thermal Engineering Technology
- BACHELORS OF HEALTH SCIENCE (B-HES)
 - Physiotherapy
 - Medical Laboratory
 - Nursing Science
- LIPRO
 - Genie Civil
- MPRO

➤ Genie Civil

2. *L'institut Supérieur des Technologies et du Design Industriel (ISTDI)*

It consist of :

❖ **BTS PROGRAMS**

- Génie Chimique des procédés (GCP)
- Maintenance des systèmes informatique (MSI)
- Informatique Industriel (II)
- Electrotechnique (ET)
- Electronique (EN)
- Froid et climatisation (FC)
- Maintenance et après-vente automobile (MAVA)
- Bâtiment (BAT)
- Travaux publics (TP)
- Géomètre topographe (GT)
- Energie Renouvelable (ER)
- Génie Bois (GB)
- Chaudronnerie (CH)
- Fabrication Mécanique (FM)

❖ **DSEP PROGRAM :**

- Réseaux et Télécommunication (RT)

❖ **LISCENCES PROFESSIONNELLES INDUSTRIELLES** in partnership with the university of DSCHANG: VIII

- Administration des systèmes Réseaux
- Génie Logiciel
- Automatique et Informatique Industriel
- Electrotechnique
- Electronique
- Management des systèmes Automobiles
- Maintenance et Expertise des Automobiles
- Maintenance des systèmes Industriels

- Génie Énergétique et Industriel
- Production et commercialisation du bois
- Génie Mécanique et productique

❖ MASTER PROFESSIONNEL COMMERCIAL in partnership with the University of DSCHANG

- Génie Electrique et informatique Industriel
- Génie des Réseaux et Télécommunications
- Systèmes d'information et de Génie Logiciel
- Systèmes d'information Audit et Conseil

3. *L'institut de Commerce et d'Ingénierie d'affaire (ICIA)*

❖ BTS PROGRAMS

- Action Commercial
- Assurance
- Banque et Finance
- Communication d'Entreprise
- Commerce International
- Gestion Logistique et Transport
- Informatique de Gestion

❖ LISCENCES PROFESSIONNELLES COMMERCIALES ET GESTION in partnership with the University of DSCHANG:

- Assurance
- Banque
- Banque – Assurance
- Comptabilité Control et Audit
- Finance – Comptabilité
- Gestion des Ressources Humaines
- Logistique et Transport
- Marketing
- Management des Opération du Commerce International IX

❖ MASTER PROFESSIONNEL COMMERCIAL in partnership with the University of DSCHANG:

- Finance – Comptabilité
- Audit Comptable et Financier
- Banque
- Fiscalité
- Management des Organisation
- Marketing – Communication
- Gestion des Ressources Humaines
- Logistique et Stratégique

❖ MASTER ISUGA – France in partnership with EMBA France

- EXECUTIVE MBA in partnership with ESSEC Douala

4. *L'institut d'Ingénierie Informatique d'Afrique Centrale (3IAC)*

❖ Cycle des TIC in partnership with CCNB-DIEPPE of CANADA

- Réseautique et sécurité
- Programmes et Analyse

❖ MASTER EUROPEAN in partnership with 3IL France

- Genie Logiciel
- Administration des Systèmes Réseaux

❖ MASTER PROFESSIONNEL in partnership with ENSP Yaoundé

- Génie Civil
- Génie Énergétique et Environnement
- Génie Industriel et Maintenance

❖ CYCLE INGENIEUR Information (3IL-France)

- Pôle de Recherche Innovation et Entrepreneuriat

5. LES PROGRAMMES INTERNATIONAUX DES SCIENCE ET TECHNOLOGIES DE L'INNOVATION (PISTI)

- ❖ Cycle des Classes Préparatoire au Grandes Ecoles d'Ingenieurs(CP) in partnership with the University of MAINE France
 - Classes préparatoires
 - Licence Sciences et technique
- ❖ CYCLE INGENIEUR de Génie Industriel (Polytechnique-NANCY France)
- ❖ INGENIEUR BIOMEDICAL in partnership with the University of Tor Vergata (ROME II)
- ❖ ARCHITECTURE ET DESIGN INDUSTRIEL in partnership with the University of CAMERINO (ROME I)

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HND: Higher National Diploma

IUC : Institut Universitaire de la Cote **CSS:** Cascading Style Sheet

DISI : Direction de l'innovation et des system d'information.

DASI: Direction Adjoint des Service informatique

DBMS: Database Management System **OOP:** Object Oriented Programming

PHP: Hypertext Pre-processor

SDLC: System Development Life Cycle

SWE: Software Engineering

SQL: Structured Query Language

UML: Unified Modelling Language

XAMPP: Cross-platform, Apache, MySQL, PHP, and Perl

UX/UI: User Experience /User Interface.

GENERAL INTRODUCTION

The technological advancements of the 21st century have significantly transformed various aspects of our daily lives. Computers and digital technologies have become indispensable tools for accomplishing a wide range of organizational tasks. In light of this progress, we have developed a **web-based application dedicated to the management of internship processes within a corporate framework**. This platform is designed to streamline the registration and validation of interns, allowing candidates to create an account, upload their documents, and await validation from the Human Resources department.

Given the importance of efficient talent development and the central role internships play in preparing young professionals for the job market, it is crucial to modernize and digitize these processes. Our application not only facilitates registration and document verification but also integrates a **training management module**, which enables validated interns to access structured training content, track their learning progress, and complete their internship program more effectively.

By providing a reliable and accessible platform, this project aims to enhance the efficiency of HR operations, reduce paperwork, and improve the overall experience of interns within the company. In doing so, it contributes to the digital transformation efforts of organizations in Cameroon, addressing existing challenges and strengthening professional development.

To achieve this goal, this report will be structured as follows:

- **PART 1: CONCEPTUAL FRAMEWORK**
 - *Chapter 1:* Presentation of the company and its activities
 - *Chapter 2:* Activities carried out during the internship period
- **PART 2: EMPIRICAL STUDY**
 - *Chapter 3:* Diagnosis of the existing internship management system
 - *Chapter 4:* Implementation of the proposed interns' portal
- **GENERAL CONCLUSION**
- **COST ESTIMATION**
- **REFERENCES**

PART ONE: CONCEPTUAL FRAME WORK

CHAPTER ONE: GENERAL PRESENTATION OF THE COMPANY

1. PRESENTATION OF ENEO CAMEROON

A company is defined as an economic and social structure that brings together human, material, and financial resources, which are organized in such a way as to provide services adapted to clients. In this section, the company **ENEO Cameroon S.A.** Will be presented as a whole that is, its creation, its evolution, its mission, its organization, its resources, as well as its values.

1.1 HISTORY

Cameroon has experienced several developments in the distribution of electrical energy. In **1948**, the company **Energie Electrique du Cameroun (ENELCAM)** was created to manage the hydroelectric plant of Edéa I on the Sanaga River, intended to supply Douala and Edéa. In **1962**, **Electricité du Cameroun (EDC)** was established, taking charge of public electricity distribution in several towns. In **1974**, the merger of ENELCAM and EDC gave birth to **SONEL**, which absorbed **POWERCAM** in 1975. To modernize the sector, SONEL signed concession agreements with the American group **AES-SONEL**, thus allowing significant investments and improvements.

In **2014**, after the acquisition of AES shares by the British investment fund **ACTIS**, AES-SONEL became **ENEO**. Founded in 1948, ACTIS has invested in more than forty energy projects worldwide over the past ten years through its subsidiary **ACTIS Energy**, a key player in energy production, transmission, and distribution. ENEO Cameroon, headquartered at Avenue De Gaulle in Douala, is a public company employing about

3,600 people. Its staff is spread across various technical and administrative departments throughout the country.

The history of electricity in Cameroon has been marked by key stages and major transformations. As early as **1929**, the first hydroelectric plants of **LUERMANN** and **MALALE** supplied electricity in the Muyuka region. In **1946**, a public service was created to purchase most of the private plants installed by the colonists. In **1958**, the **YOKE** plant was built. Before the **1939–1945 war**, the first electricity centers of

Nkongsamba, Douala, and Yaoundé were created by the Administration and operated either by private companies or directly by the Administration itself.

In **1963**, **Electricité du Cameroun (EDC)** was created to manage all public electricity distribution, with the exception of the Edéa plants managed by ENELCAM. In **1974**, the **National Electricity Company of Cameroon (SONEL)** was founded to manage public electricity distribution, including that of former West Cameroon. In **2014**, ENEO Cameroon continued to play a crucial role in the country's energy sector, with continuous investments and constant modernization to meet the growing demand for electrical energy.

Do you want me to keep this in a **formal report style** (academic/professional) or would you prefer a **simplified English version** that is easier to read for general audiences?

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1.2. PRESENTATION OF ENEO

1. Background and Context

Eneo is a company dedicated to the production, distribution, and commercialization of electrical energy in Cameroon. Its main mission is to provide reliable and affordable electricity to its customers. To achieve this goal, Eneo strives to modernize its facilities and expand its distribution network, particularly in rural areas, in order to improve access to electricity for all.

2. Mission and Commitments

Eneo is committed to respecting environmental standards and promoting the use of renewable energies in its activities. This includes the development of electricity usage in line with the national development plans of the State of Cameroon and the provisions of its concession contract. The company also works to maintain the balance between supply and demand by significantly increasing its production capacity.

3. Responsibilities of Eneo

To ensure optimal efficiency in the production and distribution of electricity, Eneo implements continuous improvement measures. Its responsibilities include:

- Developing the use of electrical energy in accordance with the State of Cameroon's development plans and concession contract provisions.
- Maintaining the balance between supply and demand by increasing production capacity.
- Continuously improving efficiency in electricity production and distribution.
- Providing quality electricity at competitive tariffs, while ensuring optimal service to its customers.

4. Strategic Vision

Through these efforts, Eneo aims to become a key player in Cameroon's energy sector by offering sustainable and innovative solutions to meet the country's growing energy needs.

1.3. Legal Form of Eneo Cameroon

Eneo is a **public limited company (PLC / Société Anonyme - SA)** with private capital, distributed as follows:

- **51%** owned by the British group **Actis**
- **44%** owned by the **Government of Cameroon**
- **5%** owned by the employees, who are organized under the **Common Initiative Group for the Socio-Economic Promotion of Personnel (GIC-P)** and affiliated with three trade union organizations.

1.4. Organizational Structure of the Company

To effectively achieve its objectives, **Eneo Cameroon** is structured into several departments and directorates. The organizational chart below provides a **macroscopic view** of this internal structure and illustrates how the company is functionally divided to ensure efficiency in its operations.

Design and implementation of a web platform for the integration and technical training of trainees at ENEO Cameroon.

Organigramme



Aminou Hamman Ladiye
Directeur Général

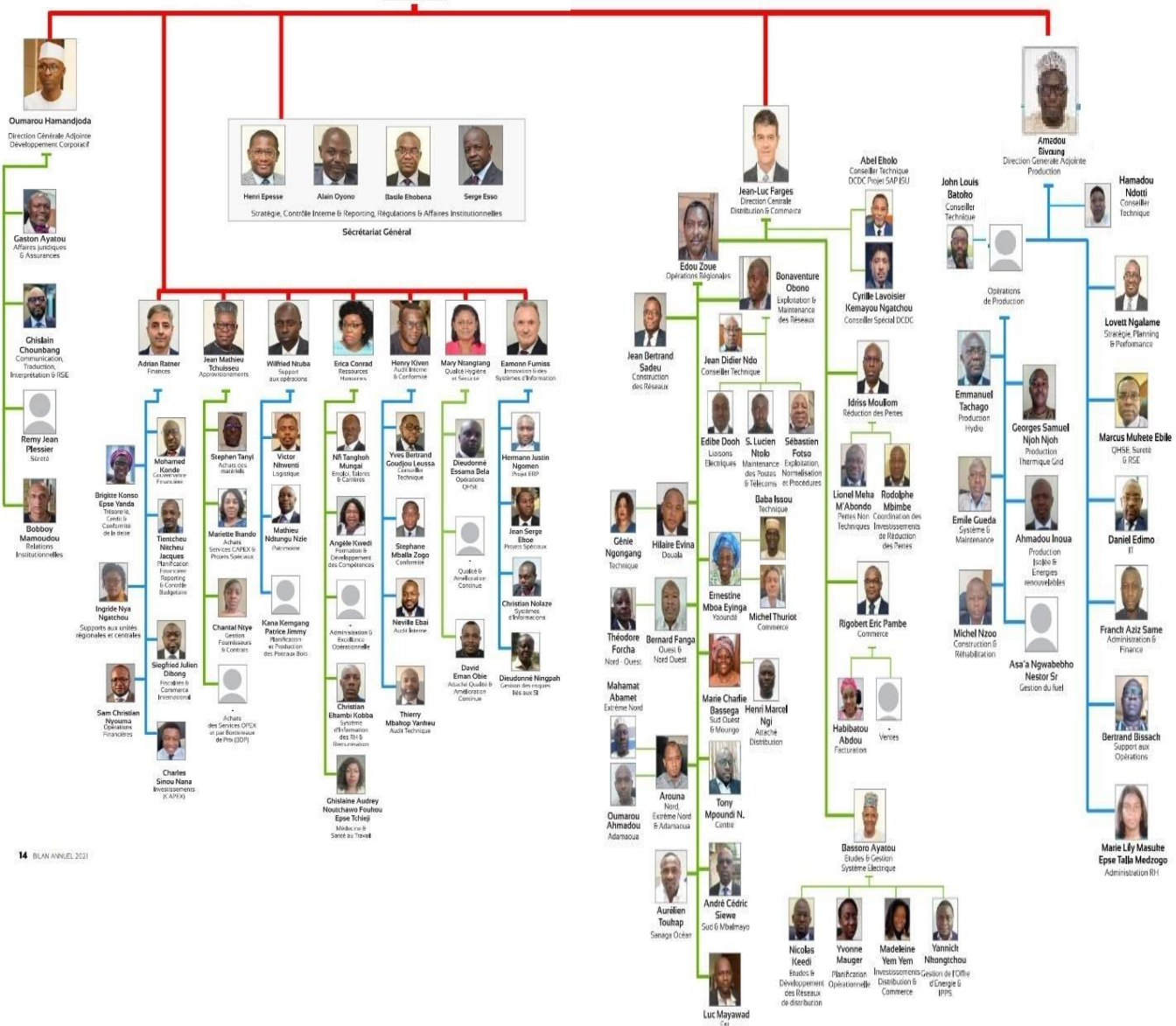


Figure 1: Organigram of Eneo Cameroon.

1.5. ENEO CAMEROON'S RESOURCES

Human Resources

Eneo Cameroon, the historic operator of the electricity sector in the country, employs approximately **3,759 staff members**. These employees are distributed across the company's core activities: **generation, distribution, and commercialization of electrical energy**.

The company places strong emphasis on continuously improving the quality of power supply while positioning itself as a model of governance in Africa. To achieve this, Eneo invests heavily in the **training and professional development** of its workforce, ensuring that employees acquire the necessary skills to address the evolving challenges of the energy sector.

Furthermore, Eneo fosters a **culture of safety, sustainability, and environmental responsibility**, implementing practices aimed at reducing its carbon footprint. The company also collaborates closely with local communities to support **development initiatives** and extend electricity access to rural areas, thereby contributing to local economic growth and improving the quality of life for populations.

Material Resources

Eneo Cameroon possesses a robust infrastructure for the **production and distribution of electricity** across the country. The company's **installed capacity is 998.939 MW**, distributed across **37 plants**—including **13 interconnected stations** and **24 isolated thermal plants**. Electricity production is derived **73.30% from hydropower**, **26.66% from thermal sources**, and **0.04% from solar energy**.

The distribution network extends over **11,450 kilometers of lines** (ranging from 5.5 to 33 kV) and **11,158 kilometers of high-voltage lines** (220 to 380 kV). By 2018, Eneo served over **1,258,340 clients**, primarily located in **Douala and Yaoundé**.

To meet the country's growing energy demand, Eneo operates **three major hydroelectric plants**—Song Loulou (384 MW), Edéa (276 MW), and Lagdo (72 MW)—as well as **three retention dams** (Mbakaou, Bamendjin, and Mapé), **six thermal plants connected to the grid**, and **26 isolated power plants**, totaling **43,016 kW**. These infrastructures enable Eneo to provide electricity to a significant portion of the national territory.

1.6. ENEO's Customer Base

Eneo Cameroon serves a **diverse customer base**, including **individuals, professionals, and industrial clients**. Individual customers primarily use the services for domestic needs, while professionals and industrial clients rely on Eneo to support their commercial and industrial activities.

The company offers **tailored solutions** such as **prepaid electricity services** and **online account management through “MyEasyLight”**, making it easier for customers to access and monitor their bills. Eneo continuously strives to **improve its services** to meet customer expectations and ensure a **reliable and efficient electricity supply**.

1.7. Company Values

Eneo Cameroon upholds **integrity, respect, commitment, and cohesion** as its core values:

- **Integrity:** Guides all actions, ensuring transparency and honesty in every operation.
- **Respect:** Promotes diversity and values different perspectives, contributing to a positive and inclusive work environment.
- **Commitment:** Demonstrated through dedication to providing high-quality service, placing the customer at the center of all priorities.
- **Cohesion:** Essential for uniting the team, fostering teamwork that supports sustainable national progress and the well-being of employees.

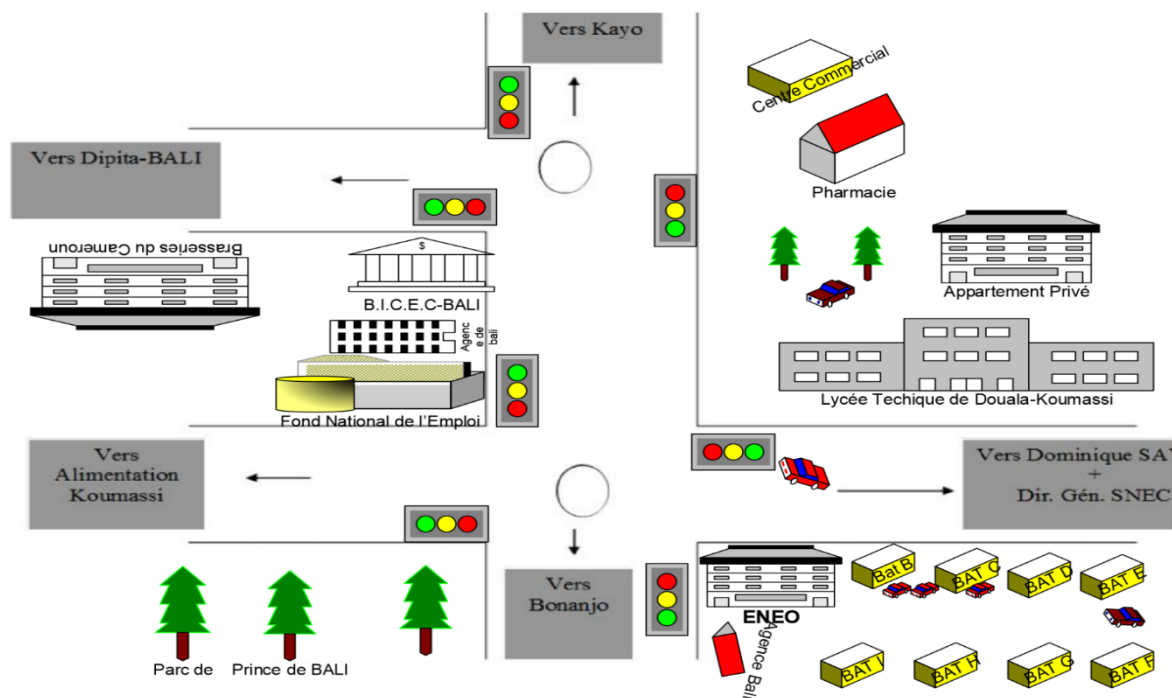


Figure 2: Geographical situation de Eneo Cameroon

2. INTERNSHIP REPPORT

2.1. Chapter 1: Activities Carried Out During the Internship

In this chapter, we will highlight the work performed throughout our internship period. We will first present the **department that hosted us** and where we carried out our activities. Then, we will describe the **specific tasks and projects** undertaken within this department, emphasizing the skills acquired and the practical knowledge gained during the internship.

Additionally, we will provide **observations, remarks, and suggestions** based on our experience, offering insights on potential improvements and best practices for the company **ENEO Cameroon**.

2.1.1. Description of the Department

Upon our arrival at **ENEO Cameroon**, we were assigned to the **Information Systems Innovation Directorate**. This department is responsible for managing both the **hardware and software** of the company's IT infrastructure, ensuring its maintenance and continuous development.

The Information Systems Directorate is composed of **four sub-units**:

- **PC Utilities Unit**
- **Service Desk (Help Desk) Unit**
- **Regional Support Units**
- **Data Center, NOC & FO Unit**

Each of these sub-units plays a critical role in supporting the company's IT operations, providing technical assistance, maintaining system reliability, and ensuring the overall efficiency of ENEO's information systems.

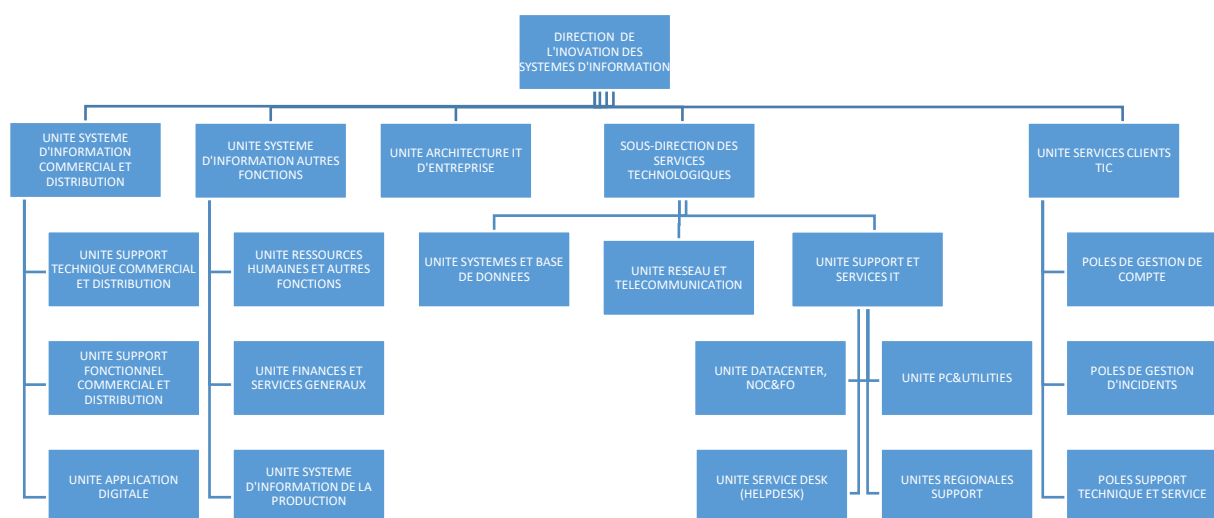


Figure 3: Organizational Chart of the Information Systems Innovation Directorate (DASI)

2.1.2. Department Responsibilities and Helpdesk Overview

The Information Systems Directorate is also responsible for ensuring the **smooth operation of the IT infrastructure**. Our internship took place specifically in the **Laboratory**, which is directly connected to another service called the **Helpdesk**.

What is a Helpdesk?

The **Helpdesk**, also known as a **Service Desk** or **Hotline**, allows a company to establish a **central point of contact** with its users. Its mission is to assist ENEO employees in **resolving IT-related issues**, whether software or hardware.

The Helpdesk is connected to both the **Laboratory** and the **IT Store**:

- **Laboratory:** Stores all defective machines awaiting repair.
- **IT Store:** Stores all new computers delivered by suppliers such as **SATI SARL, PANESS, C BUSSINNESS, INTECK, REPRO SERVICE**, etc.

User Interaction Approach

Users contact the Helpdesk to obtain solutions for their IT problems. The main channels used are:

- **Email:** Users can send a request to **ENEO Helpdesk via Outlook** or preferably create a **ticket via ITSM (IT Service Management)**, describing their issue. Once submitted, the Helpdesk sends an acknowledgment confirming receipt of the request.
- **Telephone:** Issues can also be reported via **IP-Phone**. If the problem is minor, the Helpdesk agent may suggest preliminary checks depending on the user's issue. If the problem persists, the agent or IT support intervenes to resolve it.
- **In-person:** In urgent cases, users can visit the Helpdesk directly to submit their request.

PERIOD	PERFORMED ACTIVITIES	DIFFICULTIES ENCOUNTERED	SOLUTIONS PROPOSED	SKILL ACQUIRED
From July 7 to July 11.	Introduction to company and other interns in my field	No major problem was encountered	No problem hence no solution required	Socialization skills Punctuality Team work Respect of hierarchy
From July 14 to 18	Configuration of oracle and CMS	No major problem was encountered	No problem hence no solution required	Ability to Develop theoretical Skills.
From July 21 to 25 July	Assistance to clients	No major problem was encountered	No problem hence no solution required	Ability to Apply theoretical Skills on new computers.

Design and implementation of a web platform for the integration and technical training of trainees at ENEO Cameroon.				
From 28 July to 1 August	Assistance to clients	No major problem was encountered	No problem hence no solution required	Develop Practical Skills.
From 4 august to 8 august	No major activity	No major activity hence NO problem.	No problem hence no solution required	/
From 11 August to August 15	Developing An E-commerce Platform for mentors	CMS skills was required	Learning the required skills	CMS skills acquired.
From august 18 to august 22	Developing An E-commerce Platform for mentors	No major problem was encountered	No problem hence no solution required	Improving CMS skills.
From august 25 to august 29	Assistance to clients	No major problem was encountered	No problem hence no solution required	Develop Practical Skills.

Table 1: Activities carried out during internship

2.1.2. Justification of the Topic

During my internship at ENEO, I was assigned to the Helpdesk service within the IT department. While carrying out my daily tasks, I observed that the internship application process within the company was still largely manual. Prospective interns had to be physically present to submit their documents and wait for validation, which often caused delays, inefficiencies, and unnecessary movement.

To address this issue, I proposed the development of a **web-based interns' portal**. This platform allows candidates to register online, upload their documents, and wait for validation by the Human Resources department before beginning their training.

In addition to simplifying registration, the portal also integrates a **training management module**, enabling validated interns to access training materials, follow structured programs, and track their progress throughout the internship.

This solution eliminates paperwork, reduces manual processes, and ensures a more efficient and transparent internship cycle. By engaging in this project, I was able to contribute to an initiative that enhances organizational efficiency and supports the professional development of future interns.

PART TWO: EMPERICAL STUDY

CHAPTER TWO: DIAGNOSTIC OF EXISTING SYSTEM

2.1 INTRODUCTION

An Internship Management and Training System that automates the operations of internship application, validation, and training within the Helpdesk service. This system aims to improve efficiency, accuracy, and satisfaction for both interns and HR staff.

Scope:

The system will handle:

- **Intern registration and document submission**
- **Validation and approval by HR**
- **Training content management and delivery**
- **Tracking intern progress and performance**

2.2 PRESENTATION OF THE EXISTING SYSTEM AT ENEO (HELPDESK)

Currently, the Helpdesk at Eneo Cameroon receives interns through a **manual process**. Applicants must physically come to the office with their documents (application letter, CV, academic results, etc.) for submission. The HR department manually validates these applications and contacts the selected candidates.

Once accepted, training is done orally from one technician or intern to another. This “mouth-to-mouth” transfer of knowledge is **not structured** and often leads to:

- Incomplete or incorrect training
- Miscommunication of technical procedures
- Wrong computer configurations during support, which later causes clients to return with problems

This system is inefficient, time-consuming, and prone to errors, highlighting the need for digital transformation.

2.2.1 DATA FLOW

In the **current manual system**, the data flow involves:

- **Interns (external entities)** submitting documents physically to HR
- **HR staff (process)** manually reviewing and validating
- **Helpdesk staff (process)** verbally transferring training content
- **Physical files (data store)** being kept in cabinets

In the **proposed system**, these flows will be automated through the web portal:

- **Interns** register online and upload documents
- **HR** validates applications digitally
- **Helpdesk trainers** upload structured training content
- **Interns** access training modules and track progress in real time

2.3 RESEARCH METHOD

To analyze the current system and identify its weaknesses, I applied the following methods:

- **INTERVIEW:** I discussed with HR and Helpdesk staff, who explained the internship process and challenges they face.
- **OBSERVATION:** By working alongside interns and technicians, I observed how training was delivered and where communication gaps appeared.
- **INTERNET & DOCUMENT REVIEW:** I researched best practices and modern approaches to internship management systems and training platforms.

2.4 ADVANTAGES OF THE PRESENT SYSTEM

Despite its weaknesses, the manual system has some advantages:

- **Lower Initial Costs:** No software, hardware, or IT infrastructure is required.
- **No Dependency on Technology:** The process continues even if there are network outages or system failures.

2.5 LIMITATIONS OF THE PRESENT SYSTEM

The current manual system has major limitations, including:

- **Risk of Human Errors:** Miscommunication during training leads to wrong configurations and repeat client visits.
- **Time-Consuming:** Manual document submission and oral training slow down the process.
- **Lack of Tracking:** No structured way to monitor an intern is training progress or performance.
- **Inefficient Communication:** Delays in feedback to applicants and inconsistency in knowledge transfer.

2.6 CHARACTERISTICS OF THE NEW SYSTEM

The limitations of the existing manual process highlight the need for a **new, improved Internship Management and Training System**. The proposed system will:

- **Automate Internship Applications:** Interns can register and upload documents online, reducing paperwork.
- **Enable HR Validation:** HR can review and approve applications digitally.
- **Provide Structured Training Modules:** Helpdesk staff can upload standardized training content for interns.
- **Track Intern Progress:** Supervisors can monitor completion of training tasks and performance.
- **Offer a User-Friendly Interface:** Accessible to interns, HR staff, and trainers.
- **Ensure Security:** Protect sensitive applicant data with authentication and secure storage.

CHAPTER THREE: ANALYSIS AND DESIGN OF INTERNS ENEO

CHAPTER 3: EMPIRICAL STUDY AND ANALYSIS

In order to bring forward an effective solution, a thorough analysis had to be made. The analysis phase is the first critical stage in software development. Therefore, in this chapter, I will present a complete analysis of the system I have proposed, stating the methodology I will use, requirements, design, and implementation of the system.

3.1. GENERAL PRESENTATION OF THE METHOD OF ANALYSIS

3.1.1 SYSTEM DEVELOPMENT LIFE CYCLE (SDLC)

The **System Development Life Cycle (SDLC)** is a framework used in Information Systems and Software Engineering to describe the stages of building an application. It is also referred to as the **Software Development Process**.

The cycle typically includes:

- **Planning**
- **Analysis**
- **Design**
- **Coding/Implementation**
- **Testing**
- **Deployment and Maintenance**

Different methodologies can be applied within SDLC, such as **Waterfall, Iterative, V-Model, Agile, and Prototype models**. For this project, a simplified **Agile approach** was adopted to allow flexibility and progressive improvement of the application.

3.2 PLANNING PHASE

This phase, also known as the **requirement-gathering phase**, is one of the most fundamental in SDLC. It consists of drafting a work plan and carrying out detailed feasibility and preliminary studies about what is to be done. It also involves studying the present system, identifying its weaknesses, and defining the need for improvement.

In the case of **Interns ENEO**, the existing system was **manual**. Interns had to physically visit ENEO's office to register, submit documents, and confirm their acceptance. This process was slow, time-consuming, and prone to administrative errors. Moreover, there was **no structured system to manage training, progress tracking, and validation** of interns' activities.

Therefore, I proposed the development of a **web-based application** called *Interns ENEO*, designed to:

- Allow interns to **register online** and upload required documents.
- Enable **Human Resources (HR)** to validate applications digitally.
- Provide a **training and learning portal** for interns during their internship.
- Improve transparency, efficiency, and time management in the internship process.

To collect requirements, I used the following techniques:

- **Non-structured interviews** with HR staff and IT supervisors.
- **Observation** of current manual processes at the Helpdesk.
- **Internet research** on internship management systems and best practices.

3.2.1. REQUIREMENTS GATHERING

3.2.2.1 Actors of the System

Actors are the users who interact with the system. For **Interns ENEO**, the main actors are:

- **Administrator (HR Manager / ENEO Staff):**
 - Validates interns' applications.
 - Manages training modules.
 - Tracks interns' progress.
- **Interns (Students applying for internship):**
 - Create accounts and register online.
 - Upload required documents (CV, request letter, academic transcripts, etc.).
 - Access training resources provided by ENEO.
- **Supervisors / Trainers (Optional future extension):**
 - Assign tasks.
 - Provide feedback on training progress.

3.2.2.2 Functional Requirements

The functional requirements are the specific activities and features that the system should support:

- **Intern Registration:** Online account creation and profile management.
- **Document Upload:** Submission of internship request letters, CVs, and other required documents.
- **HR Validation:** Approval or rejection of intern applications.

Identifier	User Story
REQ-1	As an admin (HR manager) , I can create and manage intern accounts.
REQ-2	As an admin , I can validate or reject internship applications.
REQ-3	As an intern , I can register online and upload my documents (CV, request letter, transcripts, etc.).
REQ-4	As an intern , I can update my personal information.
REQ-5	As an intern , I can access training materials and resources.
REQ-6	As an admin , I can assign training modules to interns.

Design and implementation of a web platform for the integration and technical training of trainees at ENEO Cameroon.	
REQ-7	As an admin , I can monitor intern progress and provide feedback.
REQ-8	As an intern , I can view the status of my application (pending, accepted, rejected).
REQ-9	As an admin , I can generate reports of intern registrations and training progress.

Additional Functional Requirements (After Refinement)

Identifier	User Story
REQ-1a	As an admin , I can delete an intern account.
REQ-1b	As an admin , I can update intern information.
REQ-2a	As an admin , I can send notifications (email/SMS) to interns about application updates.
REQ-5a	As an intern , I can download certificates or confirmations after completing training.
REQ-6a	As an admin , I can upload and update training materials.

3.2.2.3 Non-Functional Requirements

The non-functional requirements describe **how the system should perform** rather than what it should do.

Summarized Non-Functional Requirements for Interns ENEO:

Category	Requirements
Performance	The system should support up to 500 concurrent interns and HR staff without performance degradation.
Scalability	The system should scale to handle 5,000 users as ENEO expands internship opportunities.
Security	All documents and data must be encrypted using AES-256 , with secure login authentication.
Availability	The system should maintain an uptime of 99.5% , ensuring accessibility at all times.
Usability	The interface should be intuitive for both interns and HR, requiring minimal training .
Maintainability	The system should allow easy updates and feature additions without downtime.
Backup	The system should perform daily-automated backups of intern data and documents.

3.3 SYSTEM ANALYSIS

This phase serves to define the scope of the problem and set clear project boundaries. It involves breaking down the system into smaller components to better analyze the situation, identifying project goals, and determining how the proposed solution can be implemented without negatively impacting ENEO's operations. The aim is to design a solution that increases efficiency, reduces errors, and improves the intern training and management process at ENEO.

3.3.1 JUSTIFICATION OF METHODOLOGY USED

For this project, I adopted the **Agile Methodology**. Agile is particularly well suited for my project because it emphasizes **flexibility, collaboration, and iterative development**. Since the portal is intended to improve the **training and internship management process at ENEO**, requirements may evolve over time as HR managers, Helpdesk staff, and interns provide feedback. Agile allows me to adjust to these changing needs while delivering working software in incremental phases.

The Agile approach follows **seven key phases**, which I adapted for this project as follows:

1. **Requirement Gathering and Analysis**
 - I identified problems with the existing system (manual intern training and registration) through **observation, interviews with staff, and personal internship experience**.
 - Key requirements included online registration, document submission, training module access, and progress monitoring.
 - Feasibility was analyzed to ensure the project could be achieved with available time and resources.
2. **Design Phase**
 - I defined the **technical requirements** of the portal (web-based system, database for storing intern details, document management, and admin dashboard).
 - Logical designs such as **use case diagrams** and **data models** were prepared to represent the structure and flow of the system.
3. **Implementation**
 - Using web technologies, the design was translated into functional code.
 - Features such as **intern registration, document upload, training access, and admin validation** were developed in incremental steps.
4. **Testing**
 - Each feature was tested after implementation (unit testing).
 - Integration testing ensured that all modules (registration, validation, and training) worked together seamlessly.
 - User acceptance testing involved verifying that the portal met the expectations of HR staff and Helpdesk supervisors.
5. **Deployment**
 - The application was deployed in a controlled environment for demonstration.
 - Documentation was provided to explain how HR staff and interns could use the portal.
6. **Maintenance**

- Any bugs detected during deployment were corrected.
- Plans for future updates include **adding notifications, real-time chat support, and more advanced training tracking.**

7. Review or Evaluation

- Feedback was gathered from supervisors to evaluate whether the portal meets its objectives.
- This feedback will guide **future iterations** and improvements to the system.

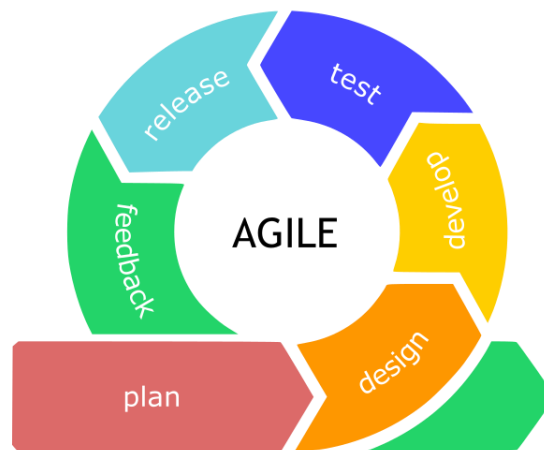


Figure 4: Agile methodology

3.3.2 Modelling Language

The modelling language I used to implement this application is the Unified Modelling Language (UML). UML is a general-purpose developmental modelling language in the field of software engineering. It provides a standardized way to visualize the design of a system, making it easier to communicate ideas and document system structure and behavior.

3.3.3 Modelling Tools

For this project, the following tools were used:

- **Draw.io (diagrams.net):** Used to create UML diagrams such as use case diagrams, sequence diagrams, class diagrams, and state diagrams.
- **Figma:** Used to design the graphical user interface (UI) templates and prototypes of the application.
- **PlantUML:** Used to generate UML diagrams (sequence diagrams, state chart diagrams, activity diagrams) from text-based descriptions for consistency and easy modification.
- **Canva:** Used for designing the application logo and other simple graphical elements.

3.4 DESIGN PHASE

3.4.1 USE CASE ANALYSIS

A use case is a list of actions or events typically defining the interactions between a role and a system to achieve a goal. The role is known in **UML** as an actor and may be human or an external system.

Actors Goal List

Initiator	Initiator's Goal	Participants	Use Case Name
Admin (HR)	Create an intern account	Database	Add Intern
Admin (HR)	Modify intern information	Database	Edit Intern Info
Admin (HR)	Delete intern account	Database	Delete Intern
Intern	Register on the portal	Database	Intern Registration
Intern	Upload internship documents	Database	Upload Documents
Admin (HR)	Validate or reject documents	Database	Validate Documents
Intern	Access training materials	Training Module	Access Training
Admin (HR)	Add or update training modules	Database	Manage Training
Intern	Update personal profile	Database	Edit Profile
Intern	Change password	Database	Reset Password

Use Case Description Examples

Use case name: Add Intern

- **Actors:** Admin (HR)
- **Actor's goal:** Register a new intern into the system
- **Preconditions:**
 - Admin has valid login credentials
 - Database is accessible
- **Main Success Scenario:**
 1. HR logs in
 2. Navigates to *Intern Management > New Intern*
 3. Fills in intern details (name, email, school, field)
 4. Saves record
- **Alternative Scenario:**
 - If required fields are missing or invalid → form is rejected
 - If intern already exists → system shows error
- **Postconditions:**
 - Intern record is stored in the database
 - Intern receives an email confirmation
- **Assumptions:** HR is logged into the portal

Use case name: Upload Documents

- **Actors:** Intern
- **Actor's goal:** Upload internship-related documents (CV, acceptance letter, ID, etc.)
- **Preconditions:**

- Intern is registered and logged into the portal
- **Main Success Scenario:**
 1. Intern logs in
 2. Navigates to *My Profile > Documents*
 3. Selects required document type
 4. Uploads file
 5. Submits for HR validation
- **Alternative Scenario:**
 - File type/size invalid → upload rejected
 - Network failure → upload not completed
- **Postconditions:**
 - Document stored in the system
 - Status updated to *Pending HR Validation*

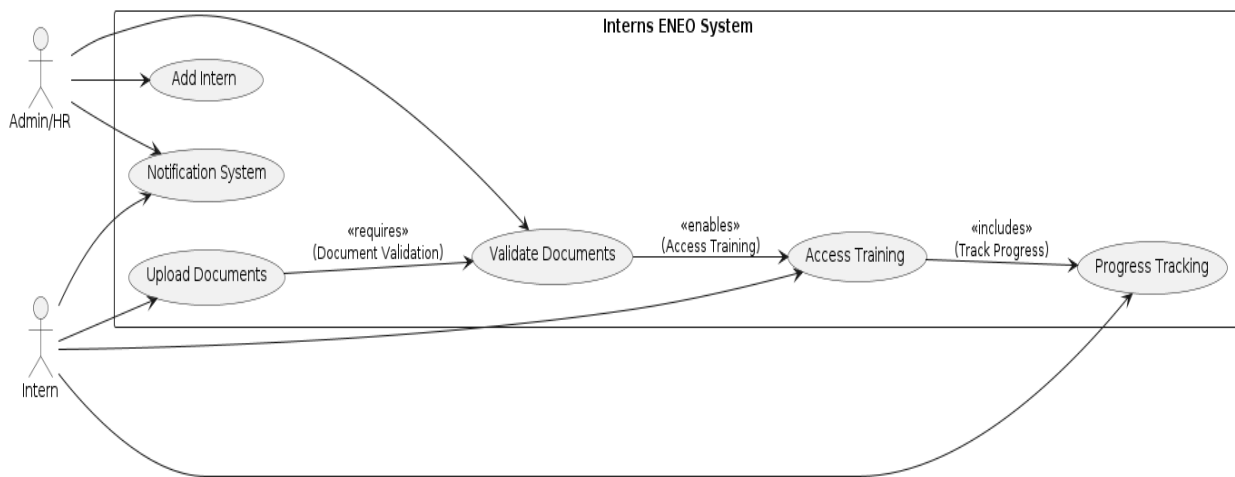
Use case name: Validate Documents

- **Actors:** Admin (HR)
- **Actor's goal:** Approve or reject submitted intern documents
- **Preconditions:**
 - Admin is logged into the system
 - Intern has uploaded documents
- **Main Success Scenario:**
 1. HR logs in
 2. Navigates to *Pending Documents*
 3. Reviews uploaded documents
 4. Approves or rejects with comments
- **Alternative Scenario:**
 - If document unreadable → reject with reason
- **Postconditions:**
 - Intern notified of validation result

Use case name: Access Training

- **Actors:** Intern
- **Actor's goal:** Access training modules once validated by HR
- **Preconditions:**
 - Intern's documents validated
- **Main Success Scenario:**
 1. Intern logs in
 2. Navigates to *Training Portal*
 3. Views available training materials (videos, PDFs, slides)
 4. Marks module as completed
- **Alternative Scenario:**
 - Training module unavailable → error message
- **Postconditions:**
 - Training progress stored in system
- **Training Portal:** Access to training materials and resources.
- **Progress Tracking:** Monitoring training status and completion.

- **Notification System:** Automatic updates to interns about application status.



Interns Eneo Use case diagram.

Figure...5

3.4.2 Sequence Diagram

A sequence diagram is an interaction diagram that details how operations are carried out, what messages are sent, and when. Sequence diagrams are organized according to time, which progresses as you go down the page. The objects involved in the operation are listed from left to right according to when they take part in the message sequence.

The purpose of a sequence diagram is to capture the behavior of the system, describe the message flow in the system, show the structural organization of objects, and illustrate interactions among those objects.

For the **Interns ENEO Portal**, the main sequence diagrams are as follows:

1. Sequence Diagram for Intern Registration

- **Actors:** Intern, System (Portal), Database
- **Flow:**
 1. Intern opens portal → clicks *Register*
 2. System displays registration form
 3. Intern fills details (name, email, school, field of study, password)
 4. System validates input
 5. System stores intern data in Database
 6. System confirms registration and sends notification email

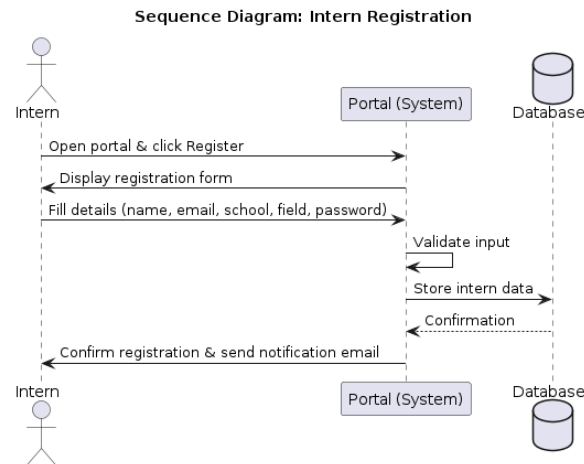
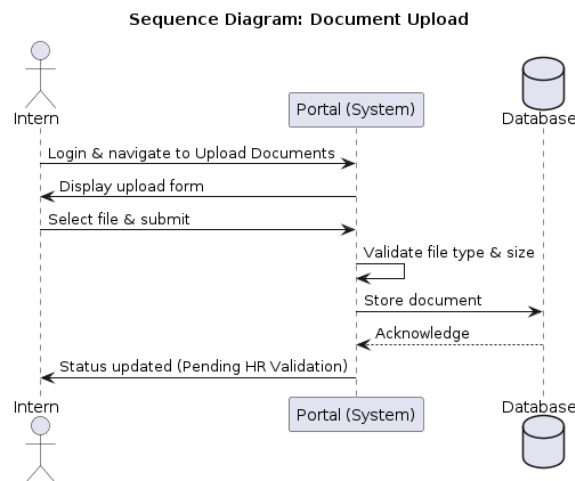


Figure 5

2. Sequence Diagram for Document Upload

- **Actors:** Intern, System, Database
- **Flow:**
 1. Intern logs in → navigates to *Upload Documents*
 2. System displays upload form
 3. Intern selects file (CV, acceptance letter, ID, etc.) and submits
 4. System validates file type & size
 5. System stores document in Database
 6. System updates status to *Pending HR Validation*



Sequence diagram: Document Upload.

Figure....6

3. Sequence Diagram for Document Validation (HR)

- **Actors:** HR Admin, System, Database, Intern
- **Flow:**
 1. HR Admin logs in → navigates to *Pending Documents*

2. System fetches uploaded documents from Database
3. HR Admin reviews documents
4. HR Admin approves or rejects with comments
5. System updates status in Database
6. System sends notification to Intern

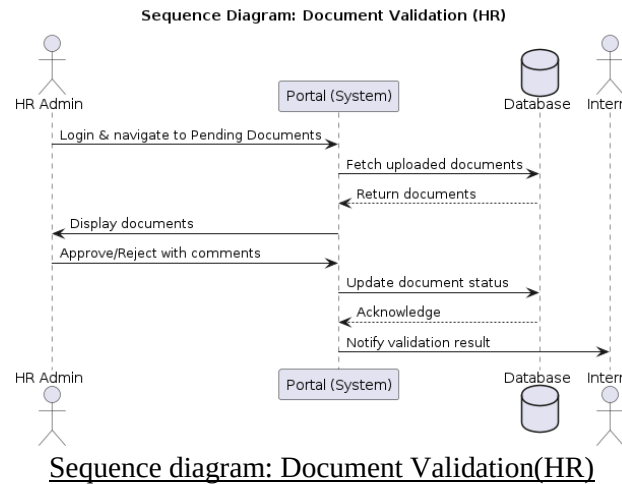


Figure...7

4. Sequence Diagram for Training Access

- **Actors:** Intern, System, Database
- **Flow:**
 1. Intern logs in → navigates to *Training Portal*
 2. System verifies that documents are validated
 3. System fetches available training modules from Database
 4. Intern selects a training module
 5. System displays content (video, PDF, slides, etc.)
 6. Intern completes module → progress saved in Database

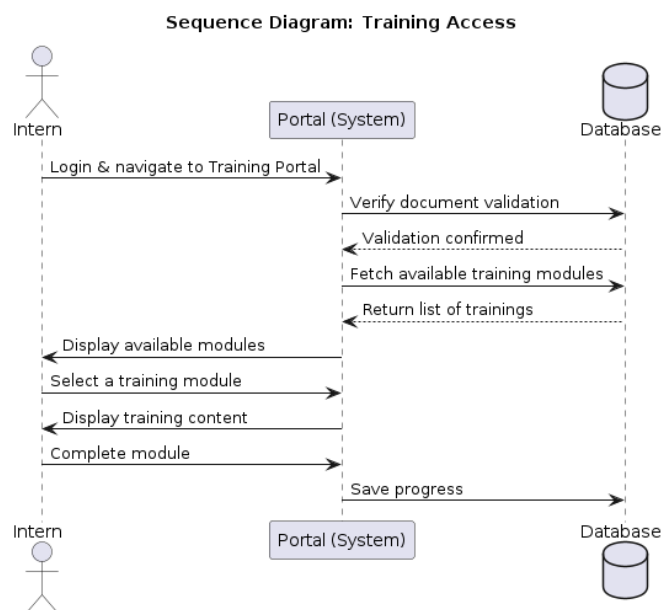


Figure...8

3.4.3 State Chart Diagram

A state chart diagram is a visual representation of the states of an object and the transitions between those states. It illustrates how an object behaves in response to various events over time. Key components include states, transitions, events, and actions. This type of diagram is commonly used in software engineering to model dynamic behavior. It aids in understanding and documenting system functionalities.

For the **Interns ENEO application**, the state chart diagram was used to represent the different states an intern or administrator can go through when interacting with the system (e.g., login, dashboard access, submit report, approve/reject report, logout).

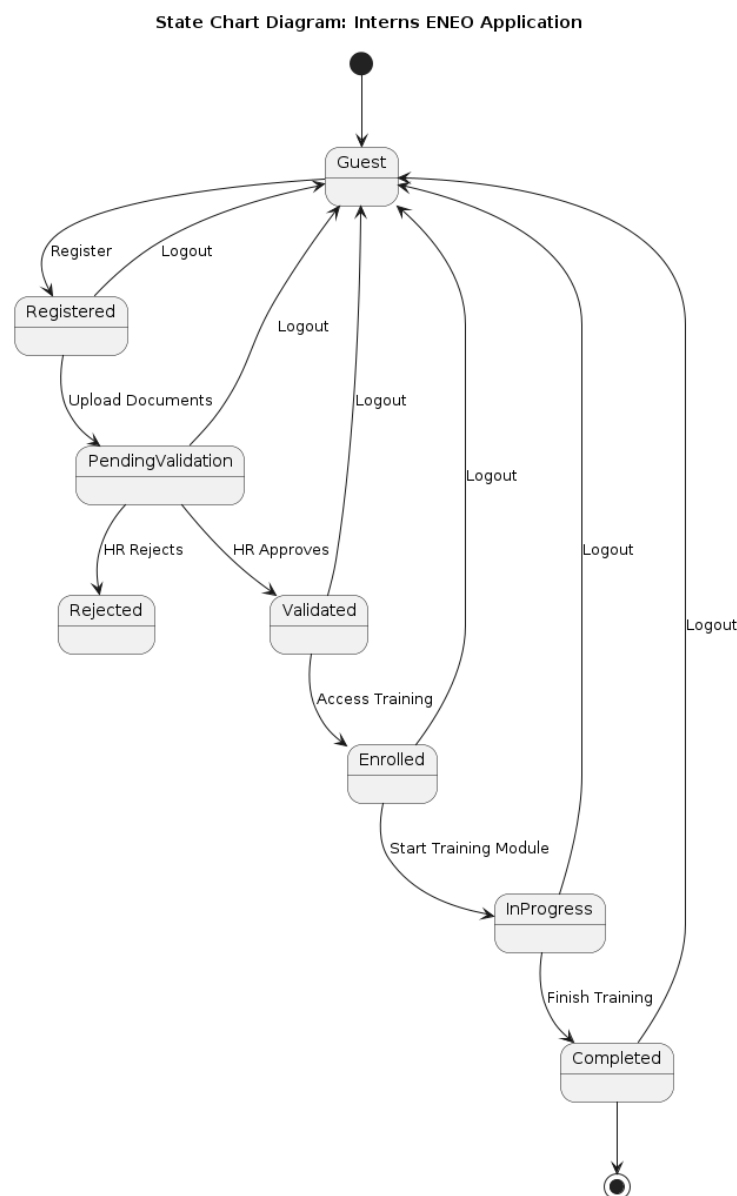


Figure9

3.4.4 Deployment Diagram

A deployment diagram is a UML diagram that illustrates the physical architecture of the system. It shows how software artifacts are deployed across hardware nodes and how these nodes interact through a network.

For the **Interns ENEO System**, the deployment diagram represents how the web application and its components (frontend, backend, and database) are distributed across different servers and user devices. It provides a clear picture of the system's physical environment and its interactions.

Key Components in the Deployment of Interns ENEO:

- **User Devices (Nodes):**
 - Interns: Access the system via web browsers to register, upload documents, and follow training materials.
 - Administrators (HR, Helpdesk): Use web browsers to validate intern applications, manage training sessions, and monitor progress.
- **Web Server Node:**
 - Hosts the application frontend and backend (developed in a web framework such as Django, Laravel, or Node.js).
 - Responsible for processing user requests, managing authentication, and coordinating training materials.
- **Database Server Node:**
 - Stores all persistent data such as intern accounts, uploaded documents, training materials, and HR validation records.
- **Network Connections:**
 - All nodes communicate through a secure network (HTTP/HTTPS).

- Encryption ensures data security during transfers.

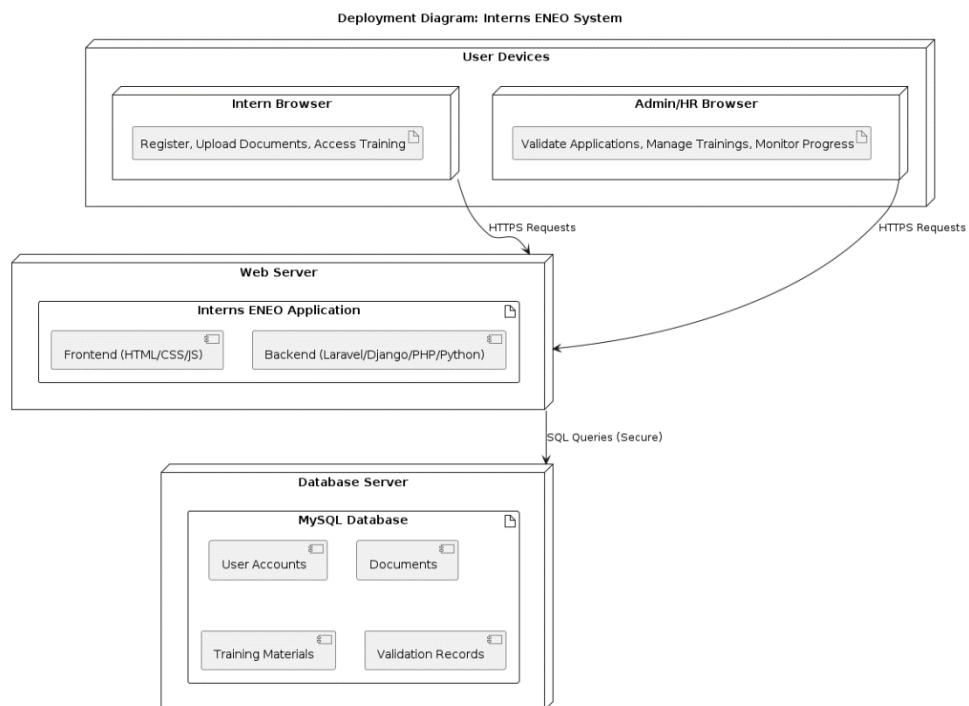


Figure...10

3.4.5 Class Diagram

A class diagram provides a static view of the system by showing its classes, attributes, methods, and the relationships between them. It helps in visualizing the structure of the system and serves as a blueprint for implementation.

In the **Interns ENEO system**, the main classes are:

- **USER**
- **ADMIN**
- **HR (Human Resources)**
- **INTERN**
- **DOCUMENT**
- **APPLICATION**
- **TRAINING**
- **NOTIFICATION**

Class Descriptions

- **USER**
 - Attributes: user_id, name, email, password, phone, role
 - Methods: login (), logout(), updateProfile()
- **ADMIN (inherits USER)**
 - Attributes: admin_id, access_level
 - Methods: manageUsers(), manageTrainings(), sendNotifications()
- **HR (inherits USER)**
 - Attributes: hr_id, department
 - Methods: validateApplication(), reviewDocuments(), approveIntern()
- **INTERN (inherits USER)**
 - Attributes: intern_id, university, study_field, status (pending/approved/rejected), training_progress
 - Methods: applyForInternship(), uploadDocument(), enrollTraining(), viewProgress()
- **DOCUMENT**
 - Attributes: document_id, type (CV, transcript, ID), file_path, status (pending/validated/rejected)
 - Methods: upload(), validate(), reject()
- **APPLICATION**
 - Attributes: application_id, submission_date, status (pending/approved/rejected)
 - Methods: submitApplication(), trackStatus(), cancelApplication()
- **TRAINING**
 - Attributes: training_id, title, description, duration, start_date, end_date
 - Methods: enrollIntern(), trackProgress(), completeTraining()
- **NOTIFICATION**
 - Attributes: notification_id, message, date, receiver_id
 - Methods: sendNotification(), markAsRead()

Class Diagram: Interns ENEO System

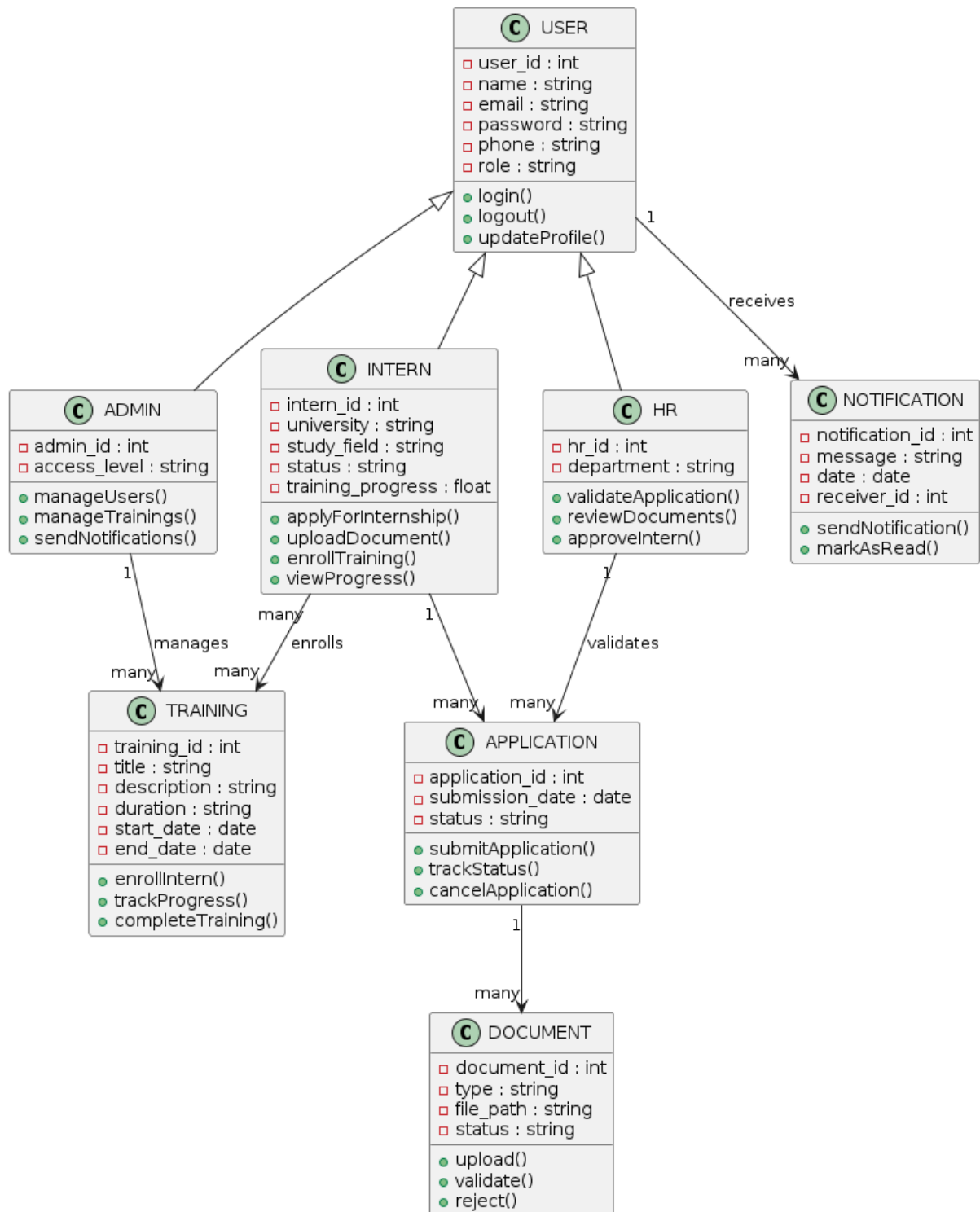


Figure...11

3.4.6 UX/UI DESIGN

The UX/UI design plays a crucial role in enhancing user experience and interaction with the **Interns ENEO web application**. A well-crafted user interface (UI) allows interns, HR staff,

and administrators to navigate through features and functionalities intuitively, making tasks more efficient and enjoyable.

The design of **Interns ENEO** emphasizes simplicity, accessibility, and responsiveness. It integrates user-friendly elements, clear navigation, and aesthetically pleasing visuals to create an engaging experience. Specific attention was given to:

- **Interns:** Easy registration process, document uploads, and training enrollment.
- **HR staff:** Streamlined validation of internship applications and tracking of interns.
- **Administrators:** Efficient management of user accounts, trainings, and system updates.

The design prioritizes **usability and accessibility**, ensuring that all users can understand and utilize the application's features without extensive training. The process involved careful consideration of user feedback and best practices in interface design, resulting in a cohesive and functional platform.

In the following section, I will present the **UX/UI mockups** created using **Figma**, which include the registration portal, dashboard, training module, and notification system.

CHAPTER FOUR: IMPLEMENTATION DETAILS AND RESULTS OBTAINED

Following the analysis phase, we moved on to the implementation of the application, which encompasses development aspects. This phase involved defining the software specifications necessary for building the application. I carefully selected the technologies and tools that would facilitate the development process, ensuring compatibility and efficiency. Additionally, I outlined the system architecture and identified any external dependencies. This comprehensive approach allowed for a smoother transition from design to implementation, setting the stage for the successful deployment of the application.

4.1. IMPLEMENTATION DETAILS

The implementation of the **Interns ENEO Web Platform** focused primarily on building a functional prototype that demonstrates how interns and administrators (HR, Helpdesk) interact with the system. The prototype simulates the core processes of the internship cycle: registration, document submission, HR validation, and access to training modules.

Although a complete deployment to ENEO's production infrastructure was not feasible within the internship timeframe, the developed prototype provides a strong foundation for future scaling and integration.

4.1.1. Frontend Development

- Implemented using **HTML, CSS, JavaScript**, with **Bootstrap** for responsiveness.
- Designed interactive pages including:
 - **Intern Portal:** Registration, login, profile management, document upload, training access.
 - **Admin/HR Dashboard:** View and validate applications, review documents, assign trainings, monitor progress.
- Interfaces designed using **Figma prototypes**, ensuring clarity, accessibility, and ease of use.

4.1.2 Backend Development

- Implemented using **PHP (Laravel framework)** for rapid prototyping.
- Core backend functionalities developed:
 - User authentication and role-based access control (Intern, HR, and Admin).
 - Document management and secure file upload.
 - Training enrollment and progress tracking.
 - Notification handling (alerts on document validation and training assignment).

4.1.3 Database Layer

- Implemented with **MySQL** for persistent data storage.

- Key tables created:
 - **Users:** stores intern, HR, and admin details.
 - **Applications:** tracks internship requests.
 - **Documents:** stores document metadata and validation status.
 - **Trainings:** training modules and schedules.
 - **Notifications:** messages sent to interns.
- Relationships between tables follow normalization standards to reduce redundancy.

4.1.4 Deployment (Prototype Stage)

- Local deployment performed using **XAMPP** for demonstration.
- Application accessible via a browser (localhost).
- Prepared for future deployment on **ENE0's intranet or cloud infrastructure** with HTTPS security.

CHAPTER V RESULTS, EVALUATION, AND FUTURE PERSPECTIVES

The Interns ENEO Portal project successfully produced a functional frontend prototype that simulates the core interactions of the platform, beginning with the Landing (Home) screen. This screen introduces the application with a clear headline and call-to-action buttons, guiding prospective interns to either register or log in. The design emphasizes corporate branding, responsiveness, and accessibility, providing users with an intuitive entry point into the system.

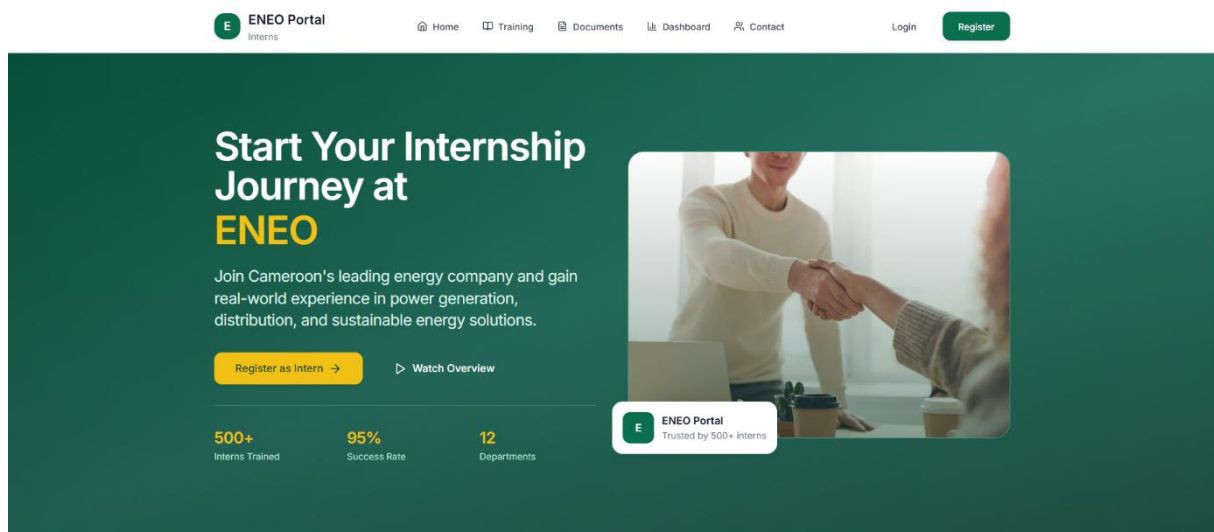


Figure 12: Landing page.

It also presents an overview of the internship process in simplified steps (Register → Upload Documents → Validation → Access Training), along with a section highlighting the benefits of joining ENEO as an intern.

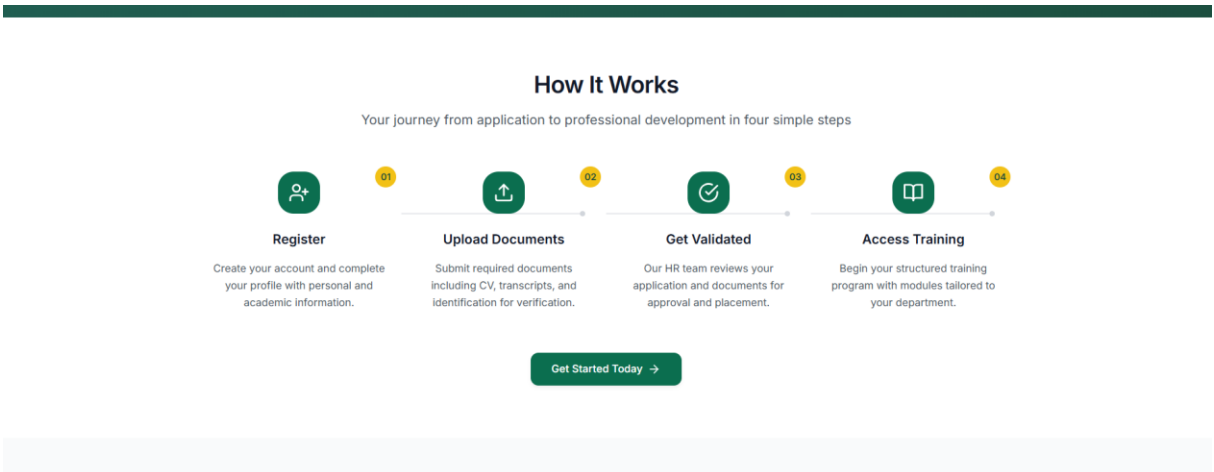


Figure 13: Presentation of the flow of work.

5.1. Key Features

On the landing screen, the *Key Features* section outlines the main advantages of the platform for interns. It highlights simplified document uploads, HR validation of applications, access to structured training modules, progress tracking, and real-time notifications. These features ensure that the internship process is transparent, organized, and easy to follow for every candidate.

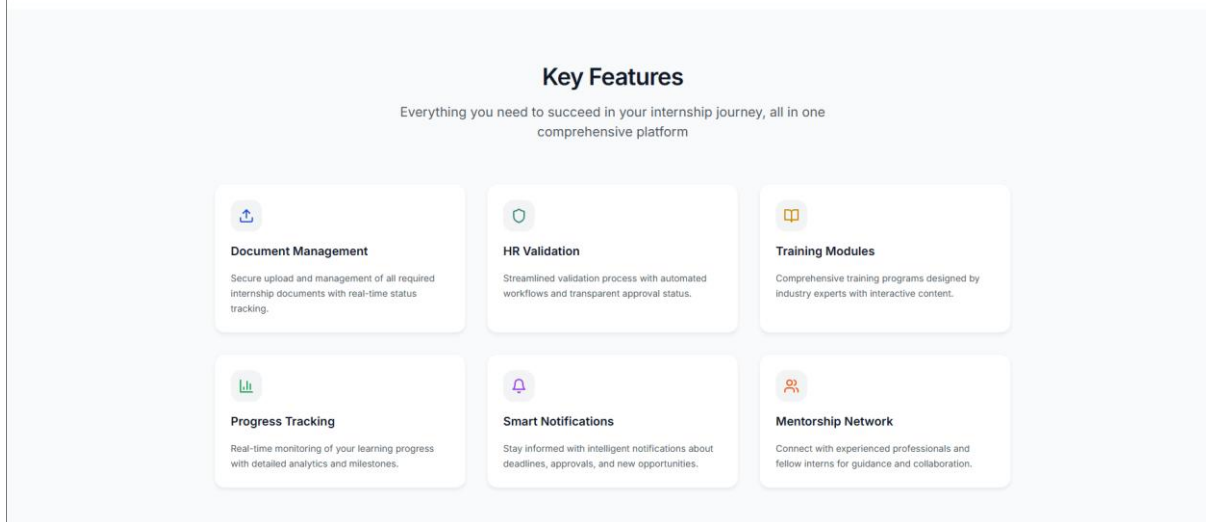


Figure 14: Key features

Why Choose ENEO for Your Internship?

ENEO Cameroon is the country's leading electricity company, committed to providing reliable power solutions and developing the next generation of energy professionals.

- Industry Leadership**
Learn from Cameroon's most trusted energy provider with decades of excellence and innovation.
- Nationwide Impact**
Contribute to projects that power communities across Cameroon and shape the country's energy future.
- Real-World Training**
Gain hands-on experience with cutting-edge technology and industry-standard practices.
- Career Growth**
Build a strong foundation for your career with mentorship from experienced professionals.

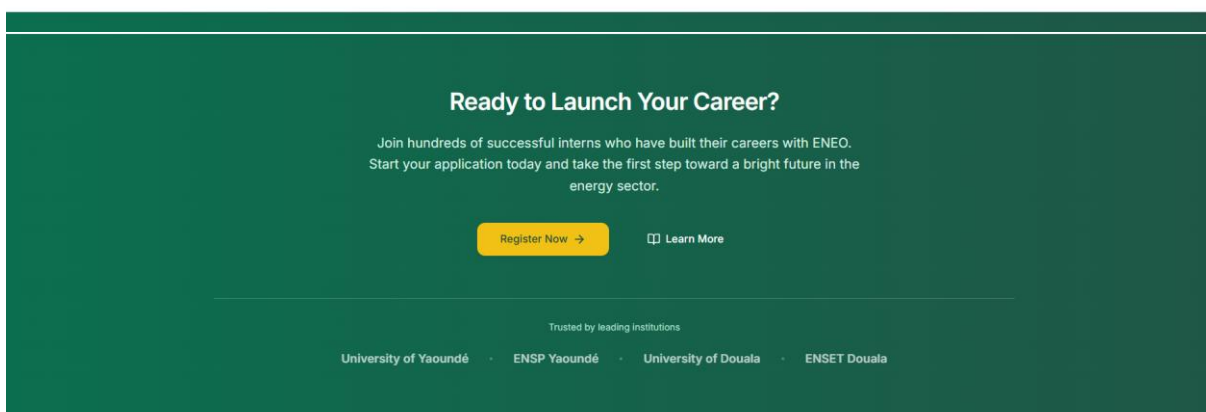
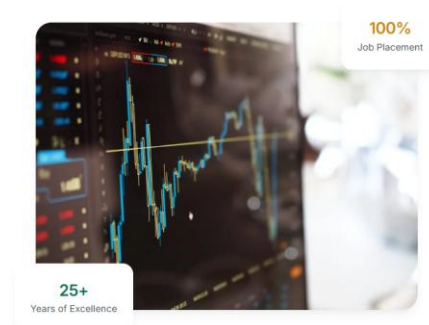


Figure 15: Why Choose Eneo.

5.2. HR Dashboard

The HR Dashboard provides an overview section that centralizes key internship metrics. From this screen, HR staff can quickly see the total number of interns registered on the platform, the number of participants who are currently completing their training, and the count of pending application or document reviews. This high-level summary offers a clear snapshot of the internship program’s progress and helps administrators prioritize their tasks.

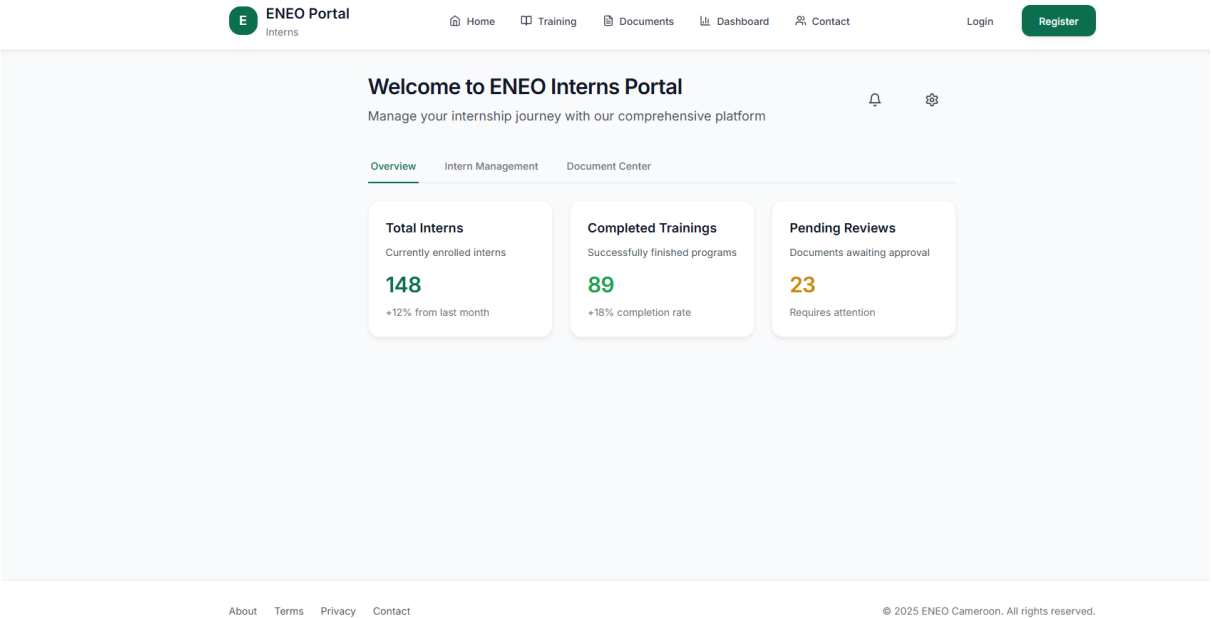
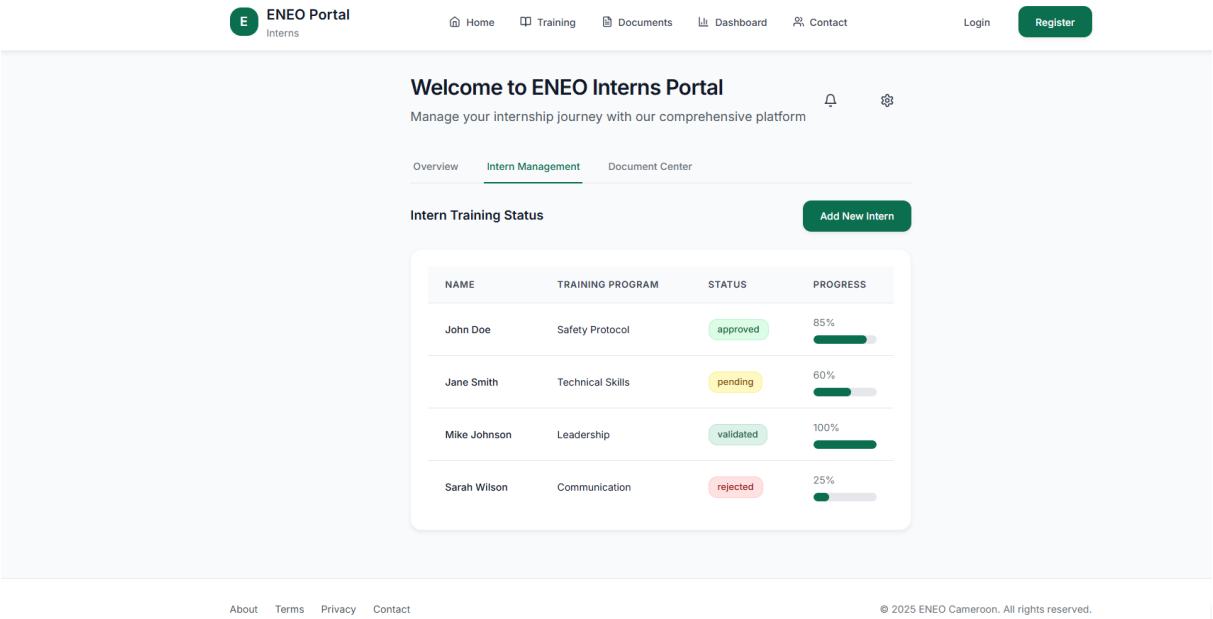


Figure 16: HR Overview Dashboard.

5.3. Interns Management

Beneath the overview, the Interns Management section offers more detailed monitoring of individual candidates. Here, each intern is listed with his or her name, assigned training program, status, and a visual progress bar that reflects his or her advancement. The interface also provides the option to add new interns. When creating a new record, a small input screen appears where the HR officer can enter the intern’s name and select their field of specialization, making the process simple and structured.



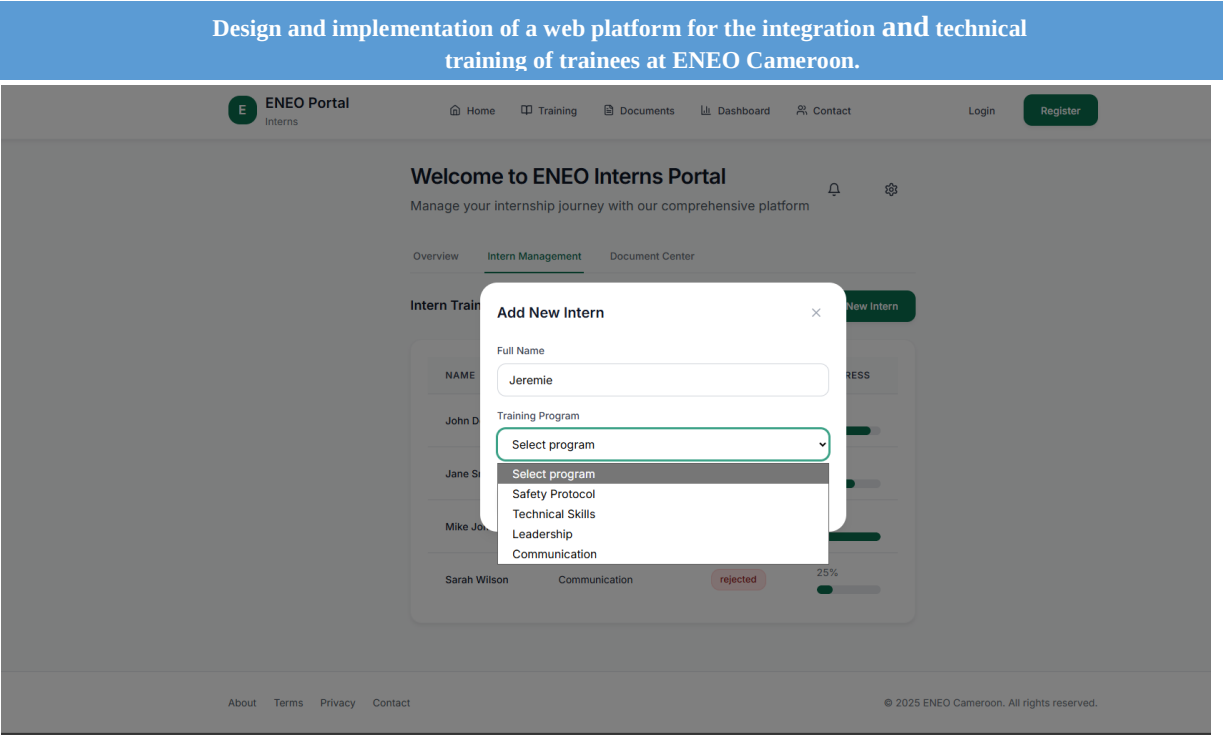


Figure 17: Interns Management.

5.4. Upload Document Center

The portal also integrates a dedicated Upload Document Center. From this screen, interns can upload required files such as CVs, identification, acceptance letters, and academic transcripts. Each document moves through a processing workflow, where its status is updated as *Pending*, *Validated*, or *Rejected*. This feature ensures transparency in the review process and gives interns a clear view of their progress toward full validation.

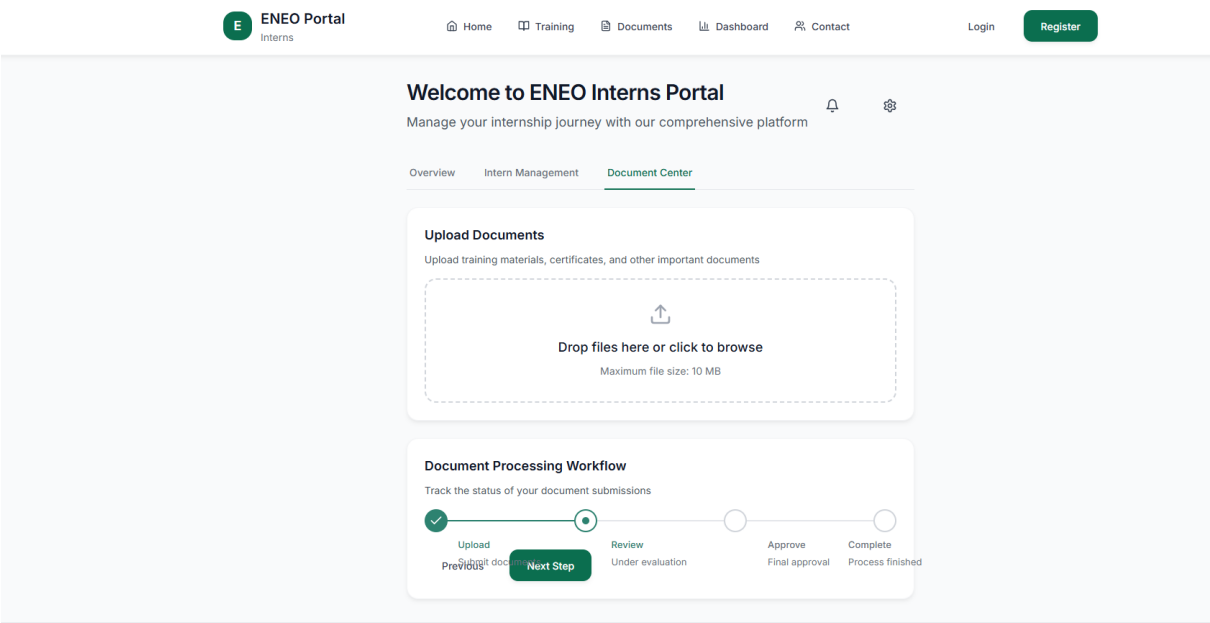


Figure 18: Upload Document Center.

GENERAL CONCLUSION

The **Interns ENEO** portal was developed to enhance and simplify the management of internships and training processes within ENEO. This system is intended to be affordable, straightforward, and user-friendly, enabling smooth registration, document submission, and training tracking for interns. The platform provides multiple features for the effective management of internship activities, from registration to completion of training programs.

Through the development of this project, I acquired significant knowledge and practical experience. Research conducted online and hands-on implementation improved my skills as a software developer, particularly in web application development, database management, and system design.

Overall, my internship at **ENEO Helpdesk** was highly constructive and enriching. The hands-on experience allowed me to apply theoretical knowledge in a real-world setting, particularly in software engineering and web development, culminating in the creation of the **Interns ENEO** portal. In conclusion, this internship was a valuable and formative experience.

PROBLEM FACED, PERFORMANCE ANALYSIS AND FUTURE IMPROVEMENT OF INTERNE'S ENEO

PERFORMANCE ANALYSIS

Given the absence of a deployed backend and real-time data services, performance evaluation focused mainly on the **responsiveness and usability** of the prototype frontend.

- **Frontend Responsiveness:** Pages and dashboards loaded smoothly using mock data. Navigation between roles (Admin, HR, and Intern) was consistent and responsive.
- **Interaction Flow Simulation:** Core workflows such as registration, login, document uploads, validation steps, and training progress tracking worked correctly under simulated conditions.
- **Scalability and Load Handling:** Could not be tested, since concurrent user handling and server scaling require a functional backend.
- **Data Accuracy:** Verification of real document validation and progress tracking was not possible, as these depend on backend logic and databases.

In summary, while technical performance under real operational loads was not measured, the prototype effectively demonstrated the intended workflows with acceptable responsiveness at the user interface level.

PROBLEMS FACED

Several limitations were identified during the course of the project:

1. **Time Constraints**
 - Full system implementation (backend services, integrations, deployment) could not be completed.
 - Only the frontend prototype with simulated interactions was delivered.
2. **Incomplete Testing**
 - System-level testing (performance, load, penetration) was not possible without backend deployment.
 - Only usability and navigation testing were performed on the frontend.
3. **Data Integration Challenges**
 - Integration with external systems such as HR databases, email/SMS notifications, or training content servers could not be achieved.
4. **Analytics & KPIs**
 - Advanced internship analytics (completion rates over time, department-level reports, etc.) could not be fully calculated without backend storage and processing.
5. **Deployment Gap**

- No cloud or on-premise deployment was achieved; demonstrations were limited to local environments within the design tool.

FUTURE IMPROVEMENTS

Despite these limitations, the prototype provides a solid foundation for further development. Future work should focus on:

1. **Backend Implementation**
 - Develop APIs for user authentication, document management, and training tracking.
 - Implement database storage for interns, HR data, and document workflows.
2. **System Integration**
 - Connect the platform to HR systems for real-time intern management.
 - Integrate with training content servers (video/PDF hosting).
 - Add notification systems (email/SMS) for status updates.
3. **Advanced Testing**
 - Perform load and stress testing to validate usability under high numbers of interns.
 - Conduct security testing to ensure safe handling of user data and documents.
4. **Enhanced Features**
 - Add dashboards for deeper analytics (training completion rates, validation timelines).
 - Include automated reminders for pending documents and tasks.
5. **Deployment & DevOps**
 - Deploy using cloud platforms (AWS, Azure, or GCP).
 - Use CI/CD pipelines for continuous updates and version control.
 - Explore containerization (Docker/Kubernetes) for scalability.
6. **User Experience Improvements**
 - Introduce a mobile-first design for easier intern access.
 - Add offline functionality for HR officers to review documents when not connected.